SIDDHI BALUJA

+91 7835979986 | siddhibaluja06@gmail.com | LinkedIn | GitHub | Certificates

EDUCATION

Amrapali University — B.Tech, Computer Science & Engineering Aug 2024 – July 2028
CGPA: 8.2/10 (till 2nd Semester) | Haldwani, Uttarakhand

Vishwa Bharati Public School — CBSE Class 12: 93% (2024) | Class 10: 96% (2022) | Noida

EXPERIENCE

Software Development Intern — Design Factory India, Noida Jun 2026 – Aug 2026
Museum & exhibition design consultancy (designfactoryindia.org) · Interaction & AI application design

- Designed and built **10+ interactive museum installations** end-to-end (concept, UX, CV/AI/IoT pipelines, kiosk deployment) for a Human & Biological Sciences Gallery — AgeWarp, Anatomy Mirror, BIONT mascot, Scale of Life, Spawning Window, Mutualism, Nucleus Experience, Prayer Plant IoT simulation, Plant Time-lapse and more.
- **AgeWarp** (ageing/de-ageing booth): fine-tuned a face-ageing model and a 6-stage image pipeline returning a 5-life-stage Polaroid strip via QR — fully offline, auto-deleting, staff-operable. **Anatomy Mirror**: touch-free gesture exhibit on MediaPipe body/hand tracking. **BIONT**: AI scientist mascot with real-time STT → LLM → TTS → phoneme lip-sync on a 3-D avatar (~2.4 s to first word).
- **Scale of Life**: 52-specimen zoom kiosk across 11.5 orders of magnitude. **Spawning Window**: WebGL coral exhibit driven by 3 physical dials. **Mutualism**: joystick-controlled projection game (Electron + Three.js). **Prayer Plant**: IoT sensor-driven plant simulation. **Plant Time-lapse**: automated capture system.

Freelance Developer, Designer & Digital Growth Consultant 2025 – Present

- Brand positioning, web design, dashboards and UI/UX for small businesses and startups; content strategy, SEO/SMO and social management for retained brands; end-to-end marketplace management across Amazon, Flipkart, Myntra and other seller panels.

PROJECTS & RESEARCH

GNSS Guardian — GPS Jamming & Spoofing-Resilient Drone Navigation **SGU HackX 1.0 — 2nd Place** · GitHub

- Raspberry Pi 4 + Quectel L86 GNSS + Pi Camera edge module that detects GPS jamming and spoofing (per-satellite C/N0 baselines, HDOP/fix and kinematic-jump checks, hysteresis state machine) and falls back to downward-camera navigation to guide the drone home.
- Optical-flow odometry (Shi–Tomasi + Lucas–Kanade, ORB keyframe breadcrumbs) at 30 fps on the Pi 4; camera-vs-GNSS cross-check flags slow spoofs; jamming detected in ~2 s on hardware; live overlay and replay dashboard.

Lunar South-Pole Ice Detection & Mission Planning — Crater F2 Bharatiya Antariksh Hackathon 2026 · GitHub

- 5-stage Python geospatial pipeline (rasterio, GDAL, NumPy, SciPy) over ~50 GB of NASA LOLA LDEM (10 m/px) and ISRO Chandrayaan-2 DFSAR L/S-band polarimetric SAR: PSR mapping, illumination modelling, ice detection, landing-site selection, rover traverse.
- Inferred ice from circular polarization ratio after showing σ^{o} inversion is degenerate for lunar ice; ice volume via Maxwell-Garnett mixing + IEM forward model; every stage self-verifies against published ground truth.

Hailmary — Multi-tenant Client Portal for Creative Studios Live · GitHub

- Next.js 15 (App Router) + TypeScript + Supabase/PostgreSQL on Vercel; passwordless magic-link and Google sign-in with admin/client roles; per-client workspaces for briefs, threads, tasks and a shared content calendar.
- Tenant isolation enforced in the database with Postgres Row-Level Security, proven by adversarial Vitest + PGlite tests; per-tenant activity feed, rate-limited export API and ZIP workspace export.

Gravity — Personal Agentic OS / Exocortex GitHub

- Python 3.11 system capturing git, browser, window and clipboard activity into a portable SQLite brain; routes natural-language Telegram messages into tasks/notes/client records and writes a daily morning brief.
- Two-plane architecture (encrypted cross-device sync for code; local-only research trail) behind a secret-redacting firewall; RAG over FTS + local embeddings, local Whisper, NVIDIA/Gemini/Groq LLM routing under a budget governor; exports a personal Obsidian vault of my own data as a linked skill graph with evidence citations.

IonWatch — Ionospheric Scintillation Monitor **IDE Bootcamp 2026 Winner · Best Performing Team · Pitch Excellence Award**

- Incubated startup: low-cost NavIC/GNSS + Raspberry Pi station computing S4 and phase scintillation indices (Solar Cycle 25) · GitHub

Sand Grain Size Mapping System **SIH 2025 — Top 5 Finalist, Ministry of Earth Sciences**

- Raspberry Pi 5 + Pi Camera system with image-processing grain segmentation and statistical analysis; 3D-printed casing.

Smart Hospitality: AI in Revenue & Yield Management Avahan 2026 National Conference

- Presented research on AI-driven pricing, demand forecasting and revenue optimisation in hospitality.

SKILLS

Languages: Python (proficient), TypeScript/JavaScript, SQL, C, HTML, CSS
Web & App: Next.js, React, Node.js, Tailwind, Supabase, PostgreSQL, Vercel, FastAPI, Electron, Three.js/WebGL, Flutter, Firebase
AI & ML: ML/DL, model fine-tuning, Stable Diffusion, Computer Vision (MediaPipe, OpenCV), NLP, LLM/RAG, STT/TTS, Generative AI
Data & Geospatial: NumPy, Pandas, SciPy, Scikit-learn, rasterio, GDAL, SAR/polarimetric analysis, DEM processing, SQLite/FTS, Jupyter
Design: UI/UX, Web Design, Figma, Canva, Interactive & Kiosk Design, Touch-free Interfaces, Design Systems, Branding, Print Layout
Tools & Practice: Git/GitHub, Vitest, E2E testing, Raspberry Pi, IoT sensors, GNSS/NavIC, Telegram Bot API, Linux/Windows deployment
Marketing & Management: SEO/SMO, Content Strategy, Meta Business Suite, Marketplace Ops, Client & Project Management

ACHIEVEMENTS

- **IDE Bootcamp 2026** (AICTE & MoE's Innovation Cell): Winner, Best Performing Team and Pitch Excellence Award for IonWatch.
- **SGU HackX 1.0** (SDGI Global University, Oct 2026): 2nd Place with GNSS Guardian; ₹11,000 prize.
- **Smart India Hackathon 2025:** Top 5 Finalist nationwide under Ministry of Earth Sciences, Government of India.
- **Startup Genesis 2025:** 2nd position at national-level startup competition; ₹5,000 cash prize.